

28	A	For half a cycle, total resistance is R. $\frac{(V_0 / \sqrt{2})^2}{R} = \frac{V_0^2}{2R}$ Power dissipation = For another consecutive half a cycle, total resistance is $(R/2)$. $\frac{(V_0 / \sqrt{2})^2}{R/2} = \frac{V_0^2}{R}$ Power dissipation = $\text{Over a complete cycle, average power dissipation} = \left(\frac{V_0^2}{2R} + \frac{V_0^2}{R} \right) / 2 = \frac{3V_0^2}{4R}$
29	A	hc